CS & IT ENGINEERING

DIGITAL LOGIC

Combinational Circuit



Lecture No. 07



By- CHANDAN SIR



01 DECODER

02 Question Practice

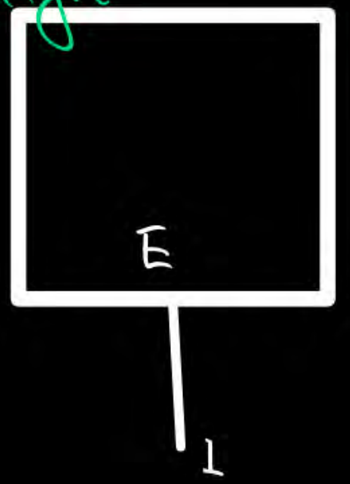
03 Discussion

DECODER

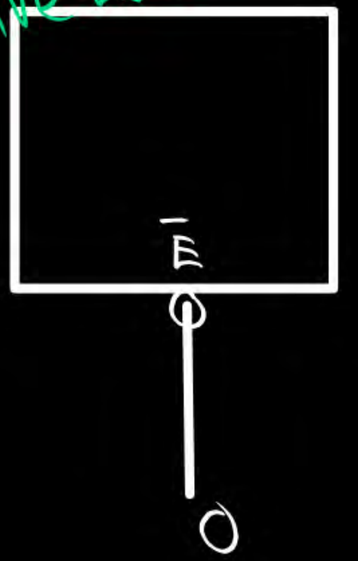
DECODER :→ Circuit which is use to convert Binary into any other code

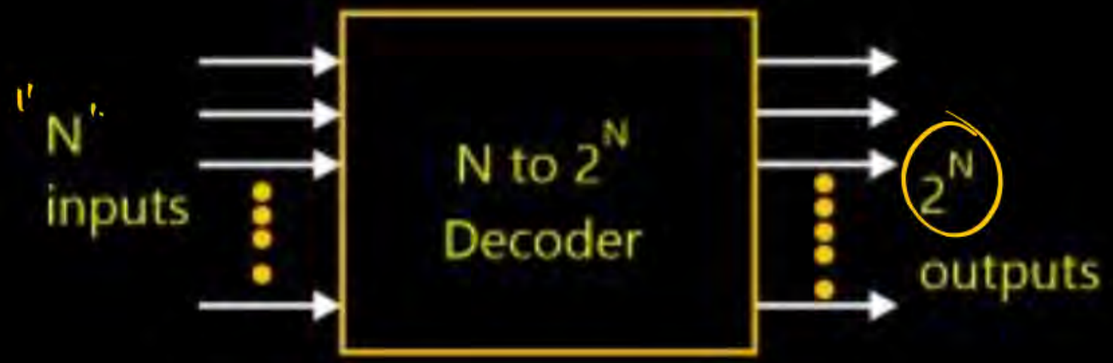
1. 2 × 4 Decoder
2. 3 × 8 Decoder
3. 4 × 16 Decoder

Active high



Active Low





DECODER

2 × 4 DECODER

Step 1.



Step 2.

MSB					LSB
I_1	I_0	y_3	y_2	y_1	y_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

DECODER

2 × 4 DECODER

$y_0 = \bar{I}_1 \bar{I}_0$

$y_1 = \bar{I}_1 I_0$

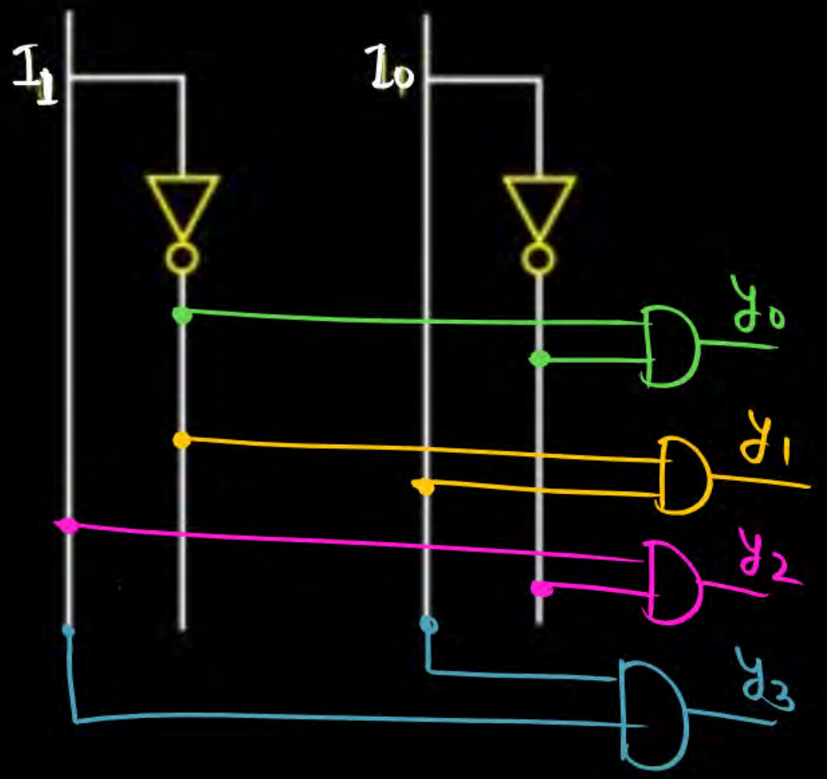
$y_2 = I_1 \bar{I}_0$

$y_3 = I_1 I_0$

Step 3.

Step 4. Minimization

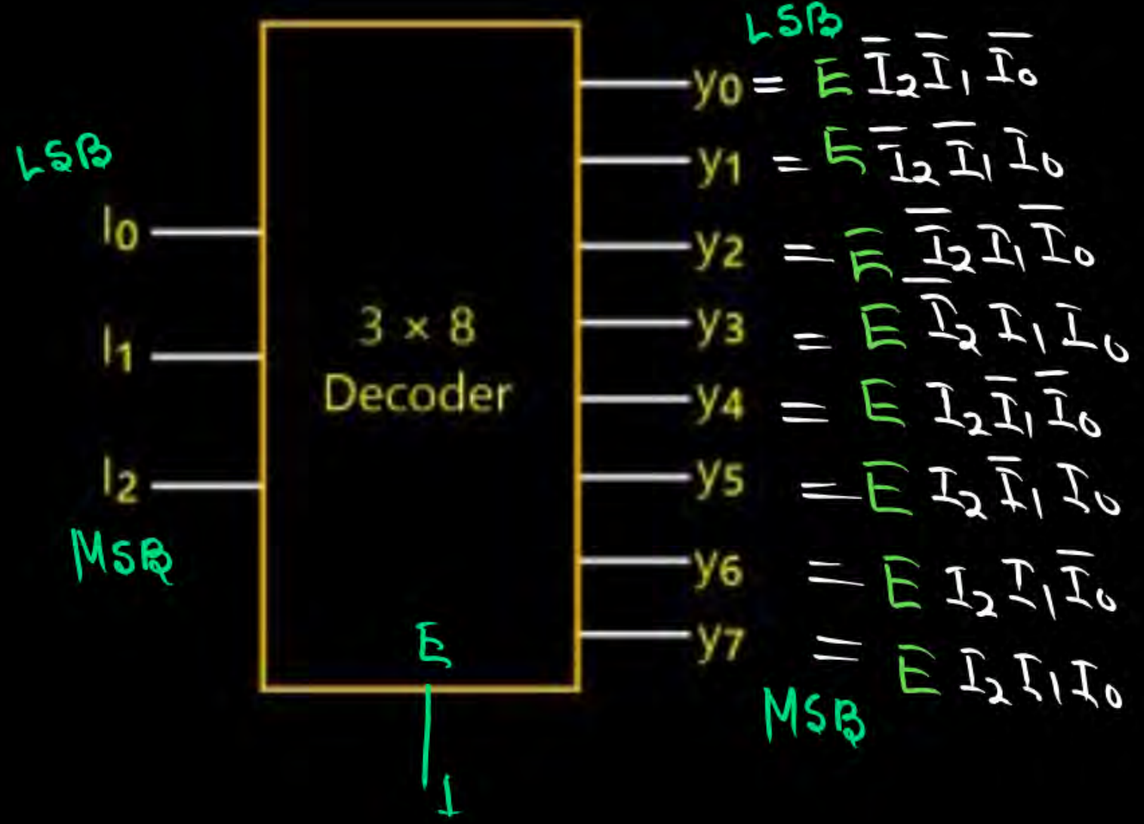
Step 5. Hardware Implementation



DECODER

3 × 8 DECODER

Step 1.



DECODER

3 × 8 DECODER

Step 2.

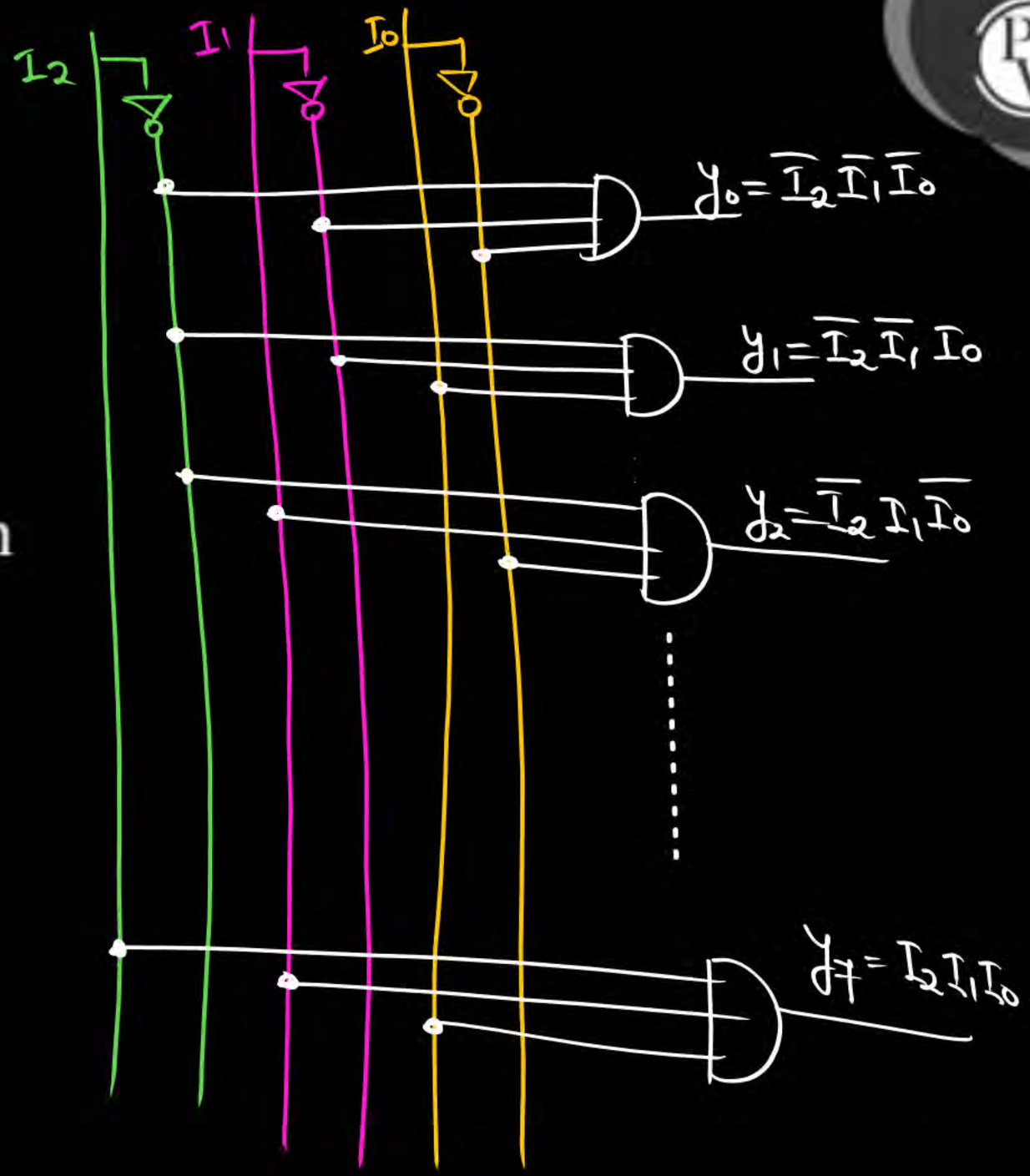
I_2 (MSB)	I_1	I_0 (LSB)	y_7 (MSB)	y_6	y_5	y_4	y_3	y_2	y_1	y_0 (LSB)
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

DECODER

3 × 8 DECODER

Step 4. Minimization

Step 5. Hardware Implementation



H.W

Design a 4X16 Decoder ?

Q Design a Binary to Decimal Decoder

input → n, output → M

$n = \log_2 M$

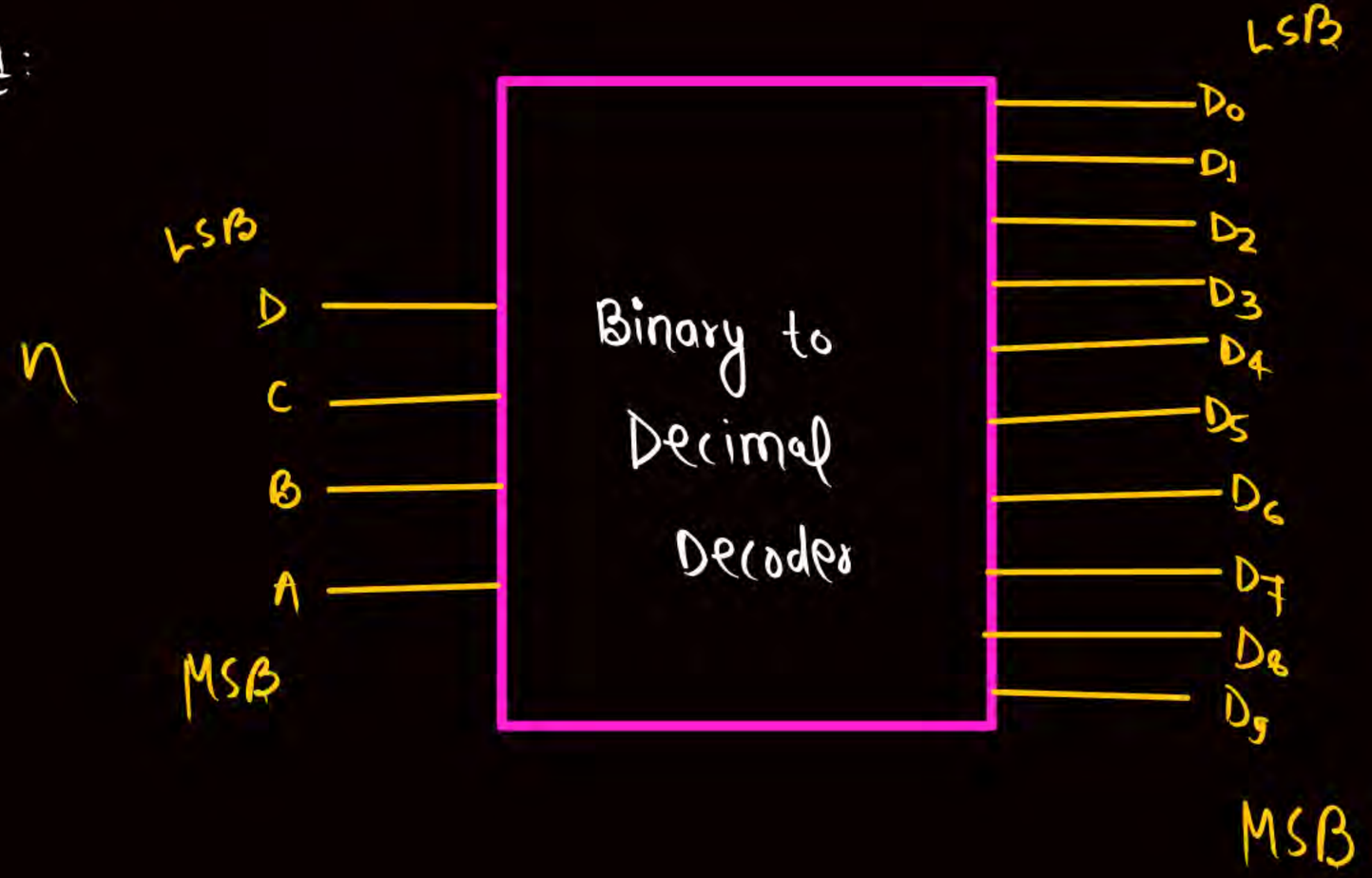
$n = \log_2 10$

$n = 3 \cdot \text{something}$

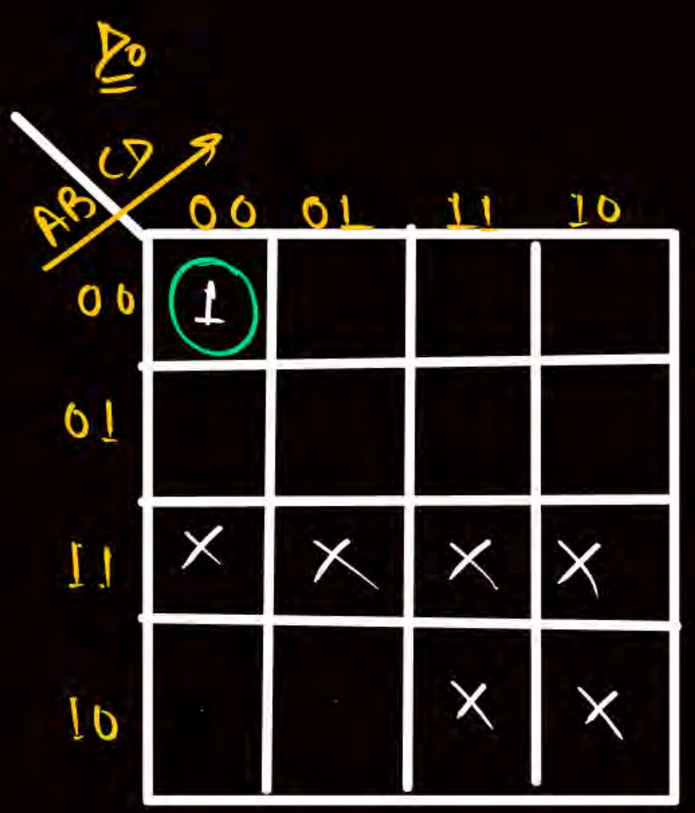
$n = 4$

$2^n = M$

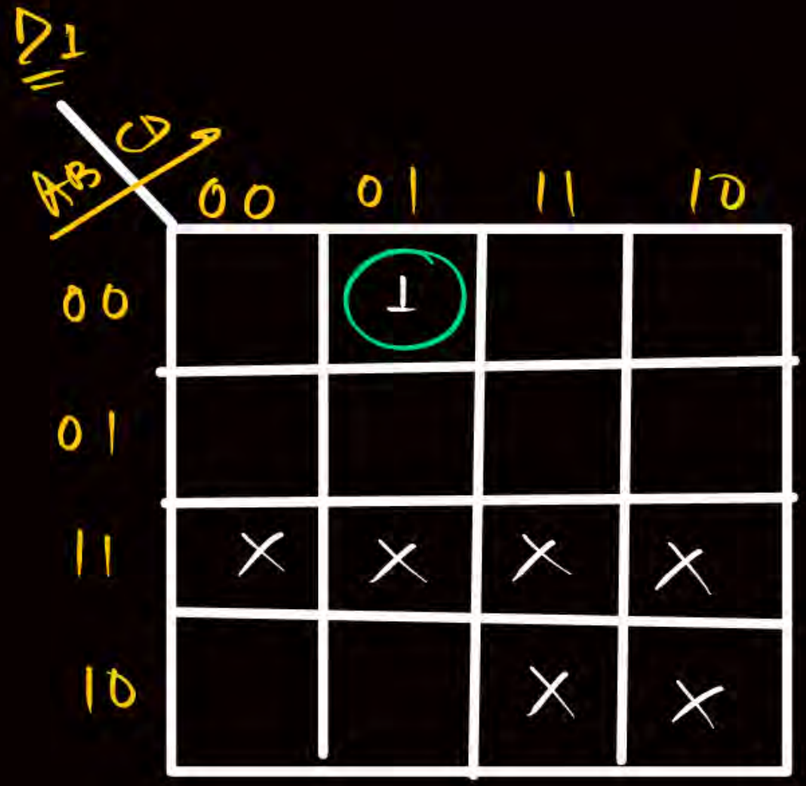
Step 1:



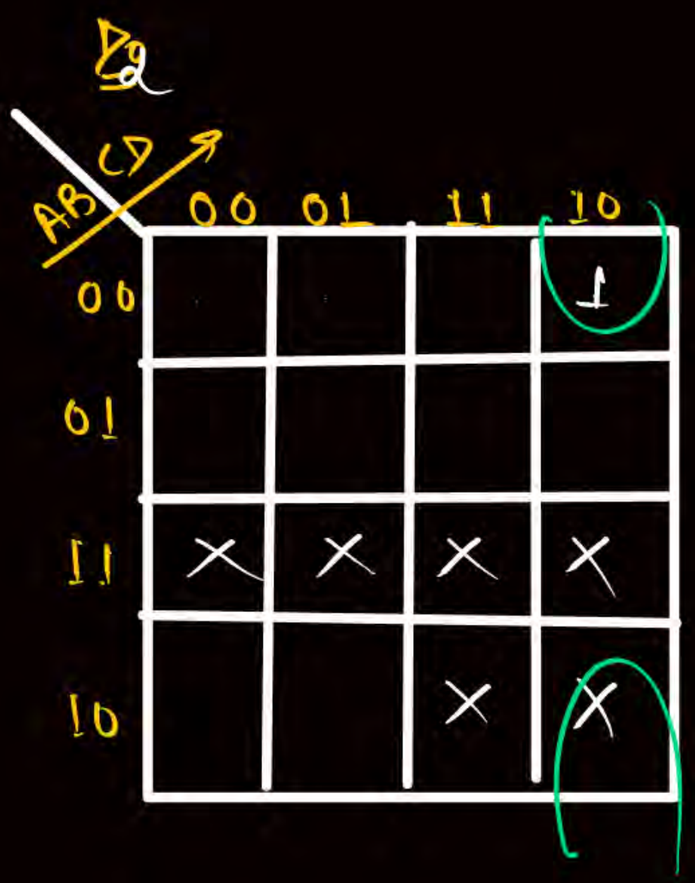
A	B	C	D	D_9	D_8	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
0	0	0	0	0	0	0	0	0	0	0	0	0	1
0	0	0	1	0	0	0	0	0	0	0	0	1	0
0	0	1	0	0	0	0	0	0	0	0	1	0	0
0	0	1	1	0	0	0	0	0	0	1	0	0	0
0	1	0	0	0	0	0	0	0	1	0	0	0	0
0	1	0	1	0	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	0	1	0	0	0	0	0	0
0	1	1	1	0	0	1	0	0	0	0	0	0	0
1	0	0	0	0	1	0	0	0	0	0	0	0	0
1	0	0	1	1	0	0	0	0	0	0	0	0	0
1	0	1	0	X	X	X	X	X	X	X	X	X	X
1	0	1	1	X	X	X	X	X	X	X	X	X	X
1	1	0	0	X	X	X	X	X	X	X	X	X	X
1	1	0	1	X	X	X	X	X	X	X	X	X	X
1	1	1	0	X	X	X	X	X	X	X	X	X	X
1	1	1	1	X	X	X	X	X	X	X	X	X	X

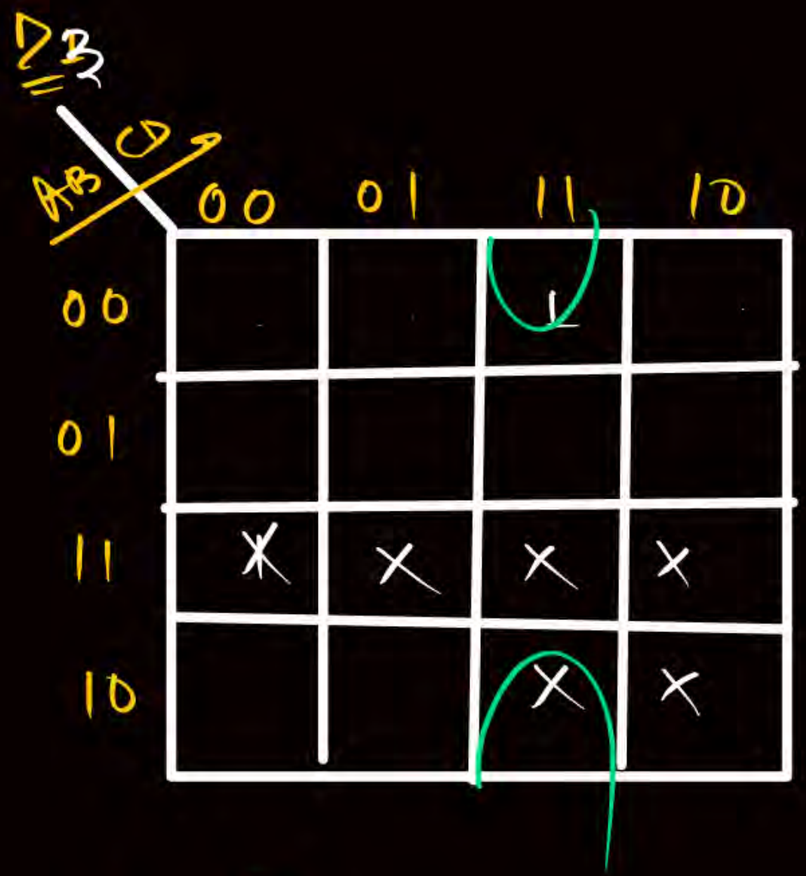


$D_0 = \bar{A}B\bar{C}\bar{D}$

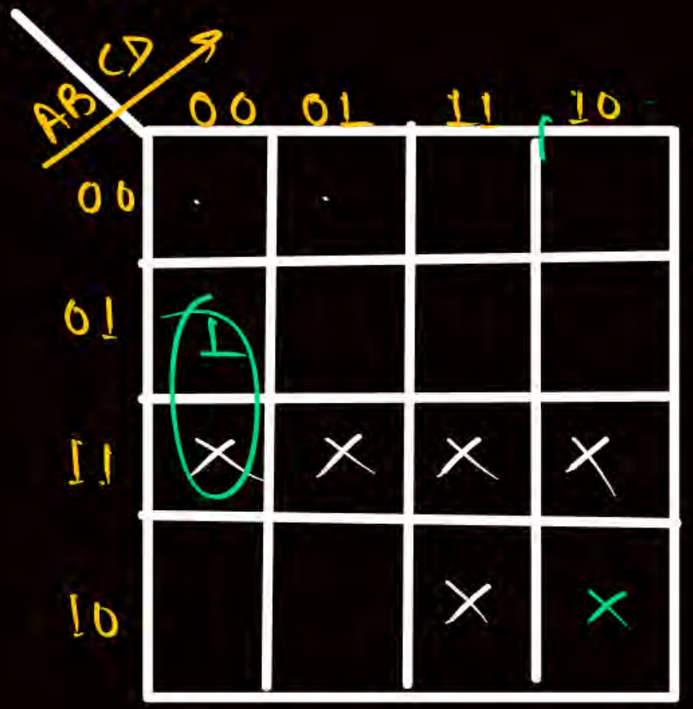


$D_1 = \bar{A}B\bar{C}D$


D_2
AB
CD
00 01 11 10
00
01
11
10
1
X X X X
X X
$D_2 = \bar{B}C\bar{D}$

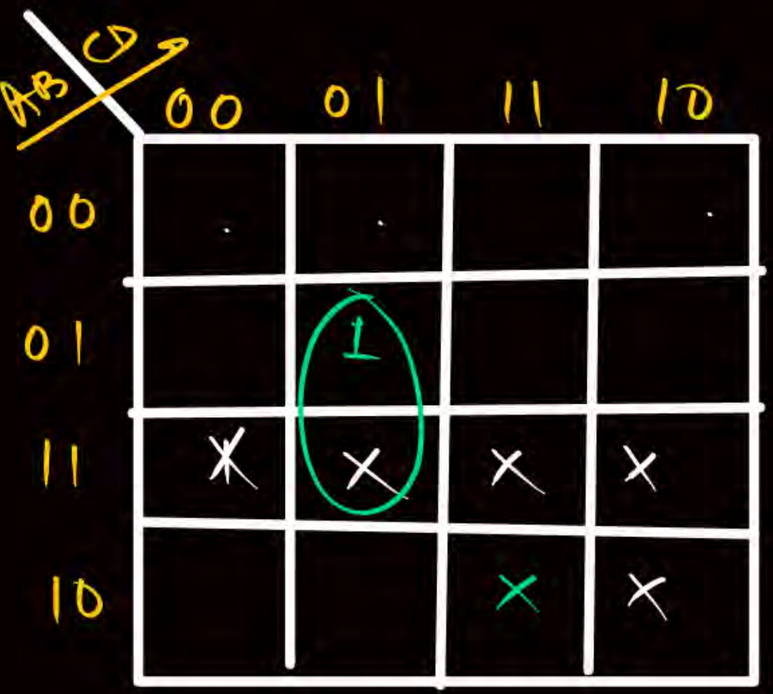

D_3
AB
CD
00 01 11 10
00
01
11
10
1
X X X X
X X
$D_3 = \bar{B}CD$

D4



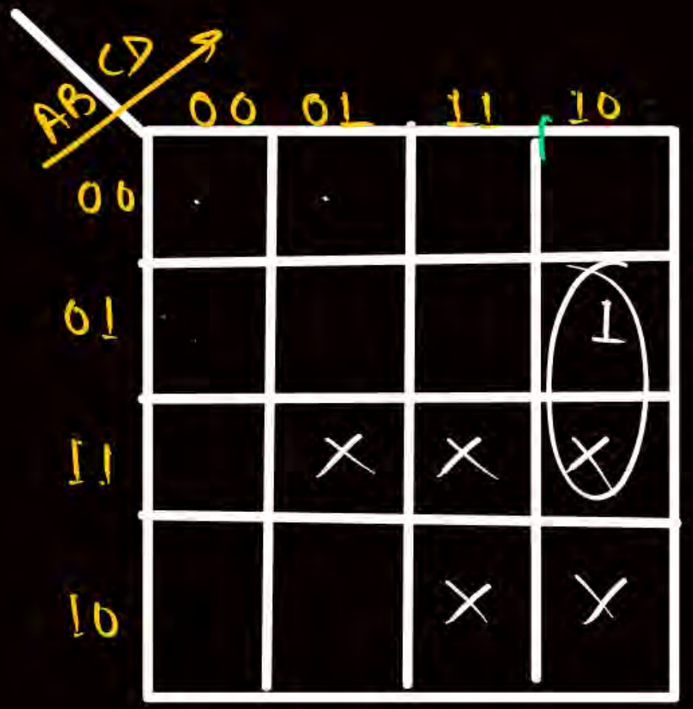
$D_4 = B\bar{C}\bar{D}$

D5



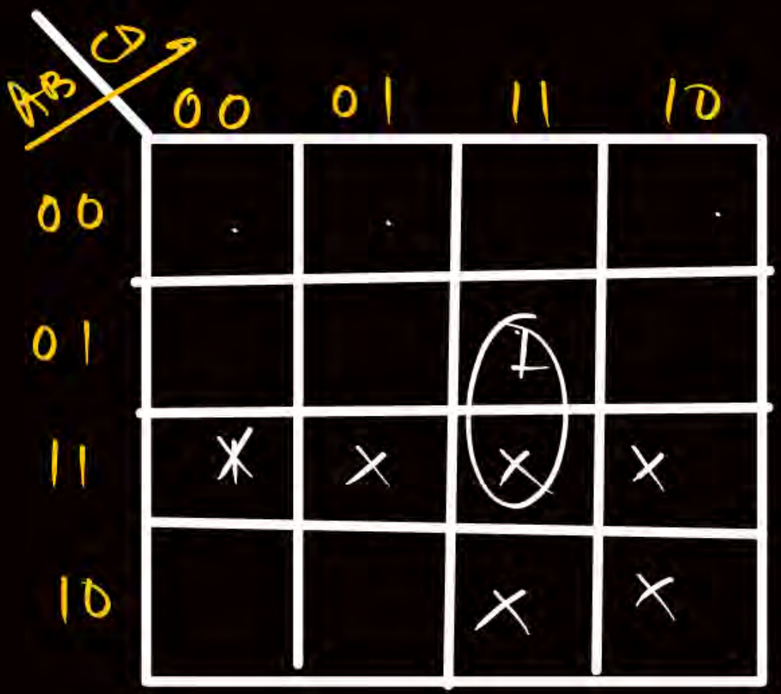
$D_5 = B\bar{C}D$

D6



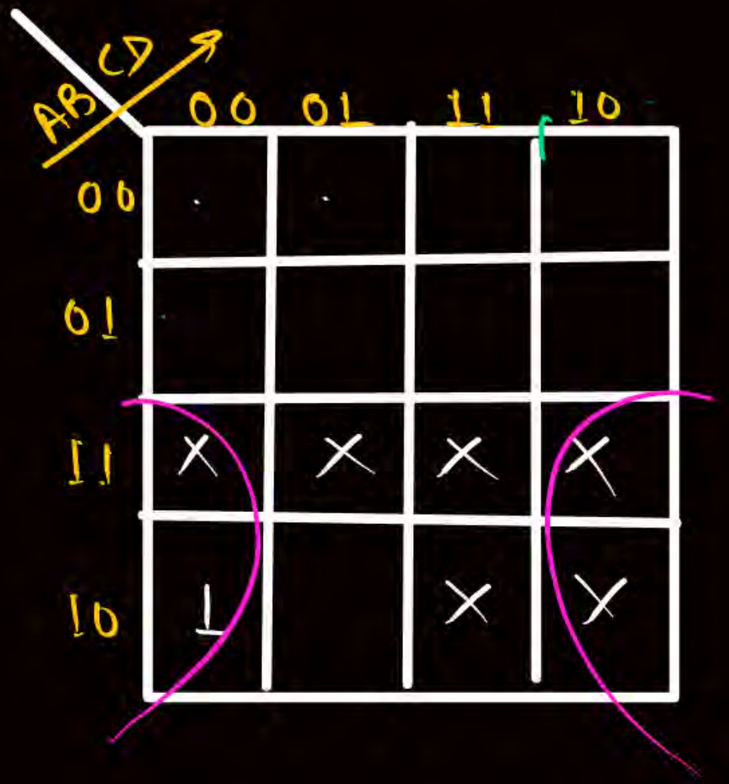
$D_6 = BC\bar{D}$

D7



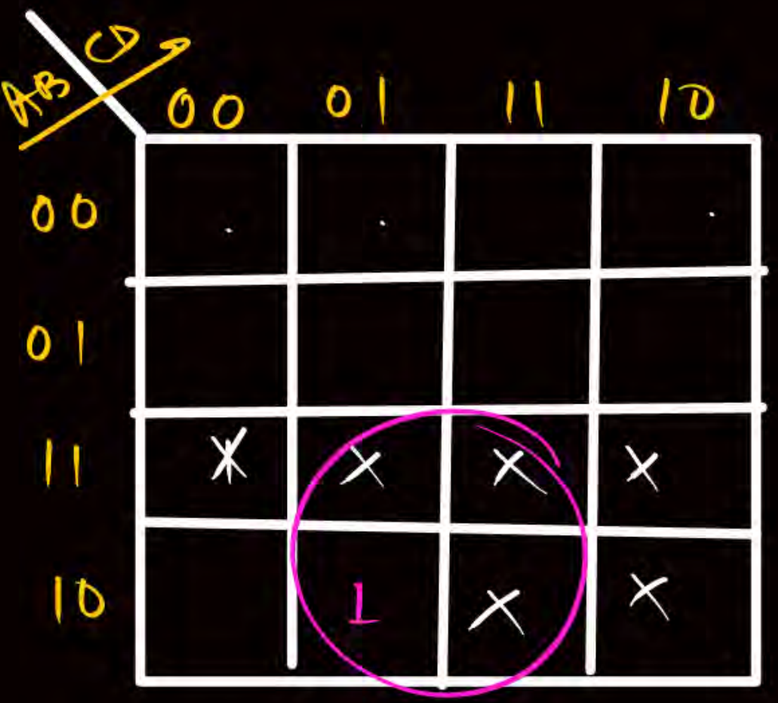
$D_7 = BCD$

D_8



$D_8 = A\bar{D}$

D_9



$D_9 = AD$



$D_0 = \bar{A}\bar{B}\bar{C}\bar{D}$

$D_1 = \bar{A}\bar{B}\bar{C}D$

$D_2 = \bar{B}C\bar{D}$

$D_3 = \bar{B}CD$

$D_4 = B\bar{C}\bar{D}$

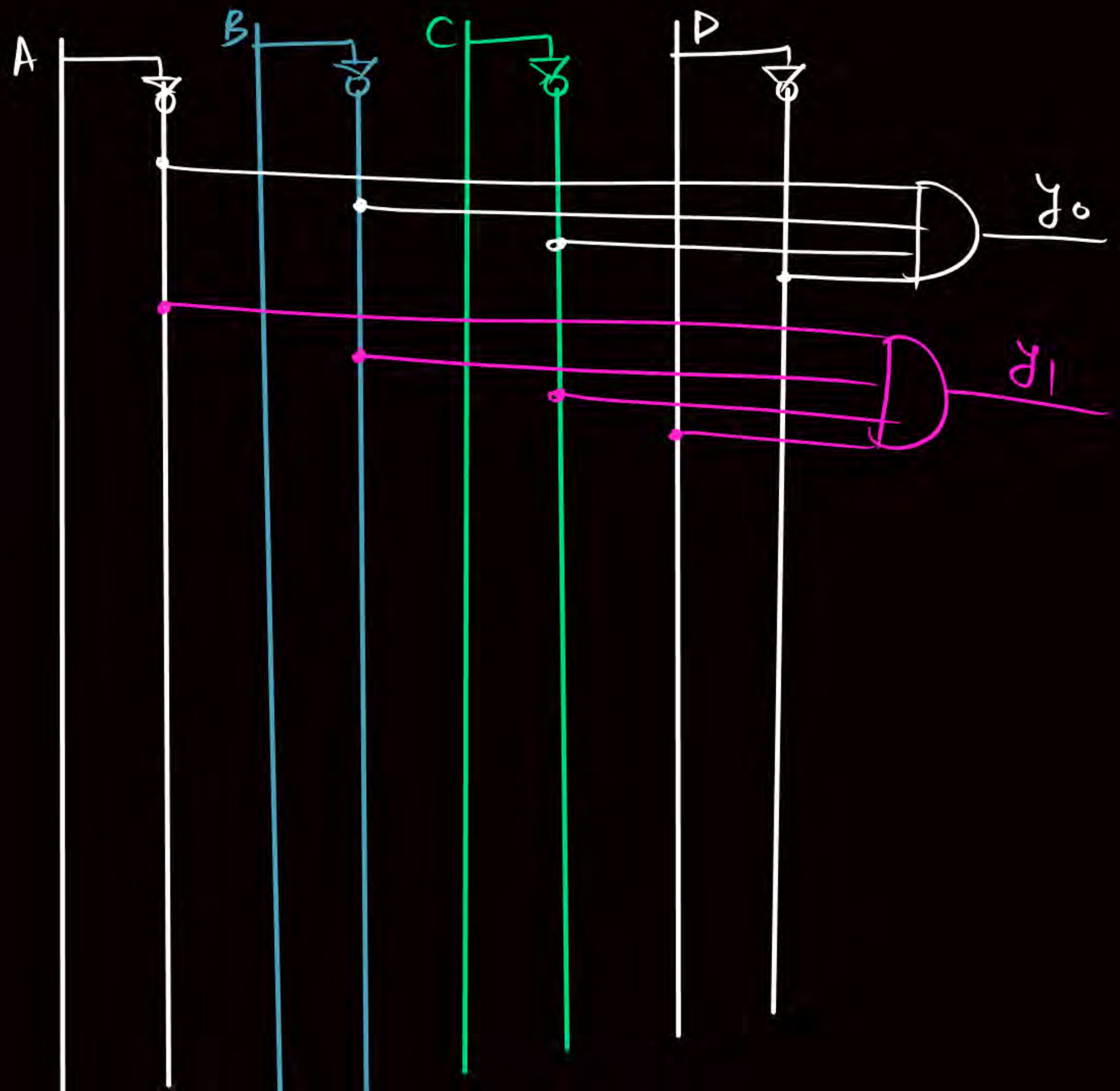
$D_5 = B\bar{C}D$

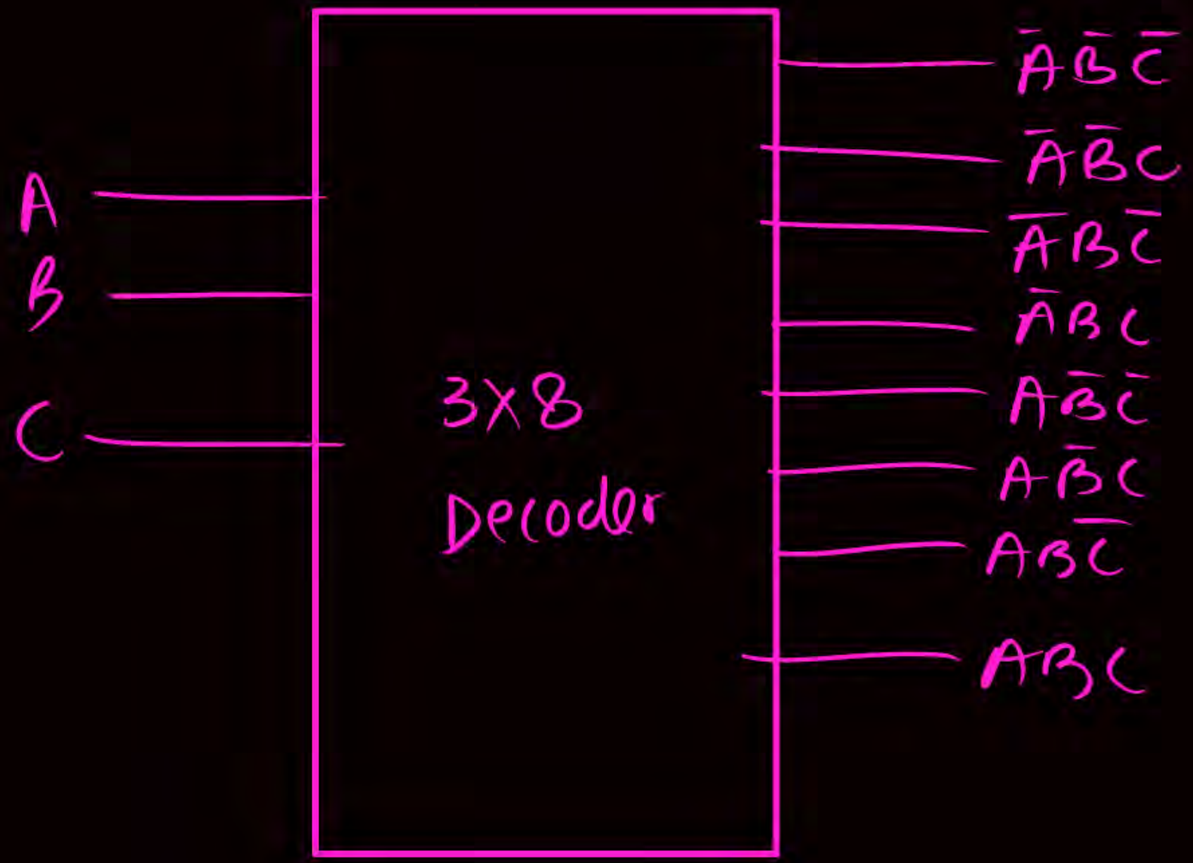
$D_6 = BC\bar{D}$

$D_7 = BCD$

$D_8 = A\bar{D}$

$D_9 = AD$


A
B
C
D
y_0
y_1


A
B
C
3X8
Decoder
$\bar{A}\bar{B}\bar{C}$
$\bar{A}\bar{B}C$
$\bar{A}B\bar{C}$
$\bar{A}BC$
$A\bar{B}\bar{C}$
$A\bar{B}C$
$AB\bar{C}$
ABC

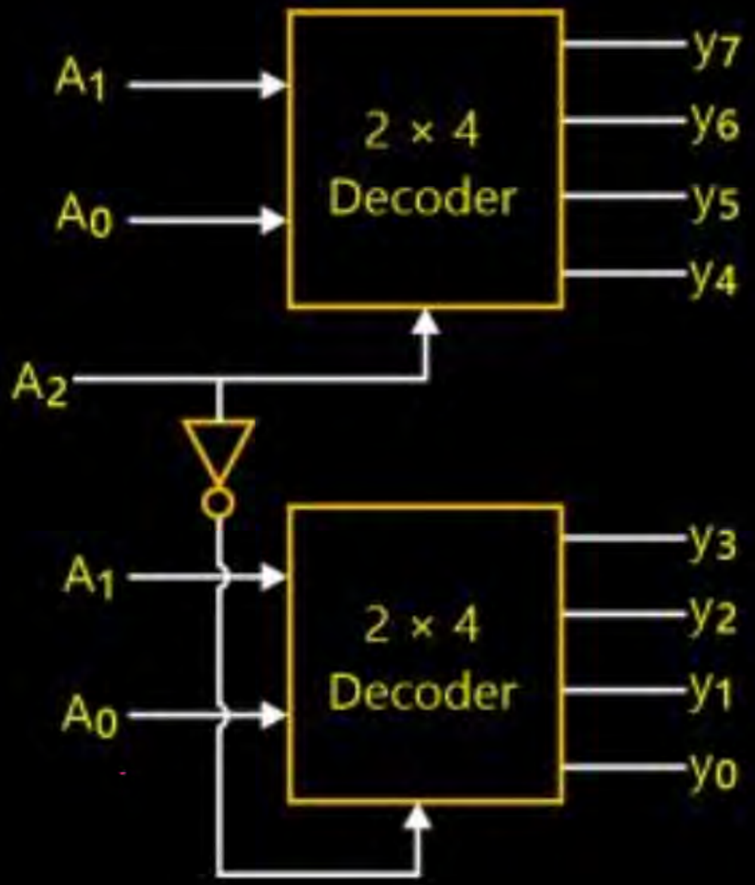
DECODER

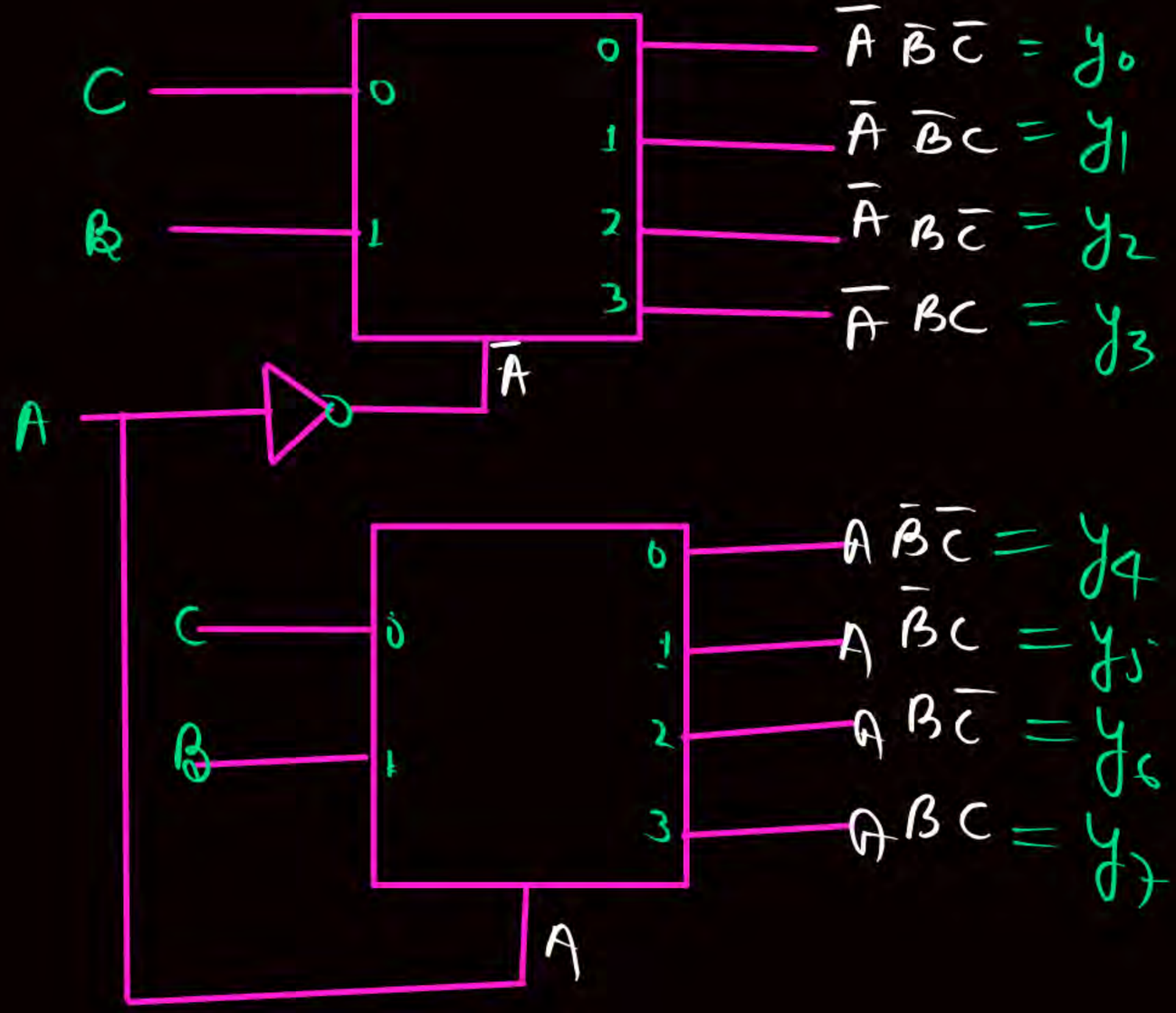
$2 \times 4 \xrightarrow{\frac{8}{4}} 3 \times 8$

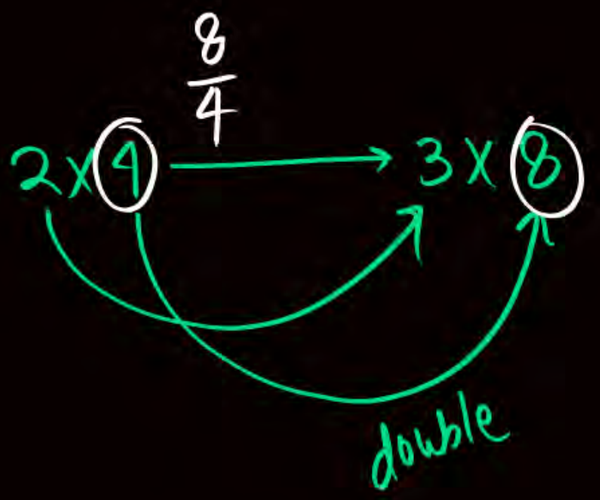
② Answer

DESIGN 3 × 8 DECODER BY USING 2 × 4 DECODER

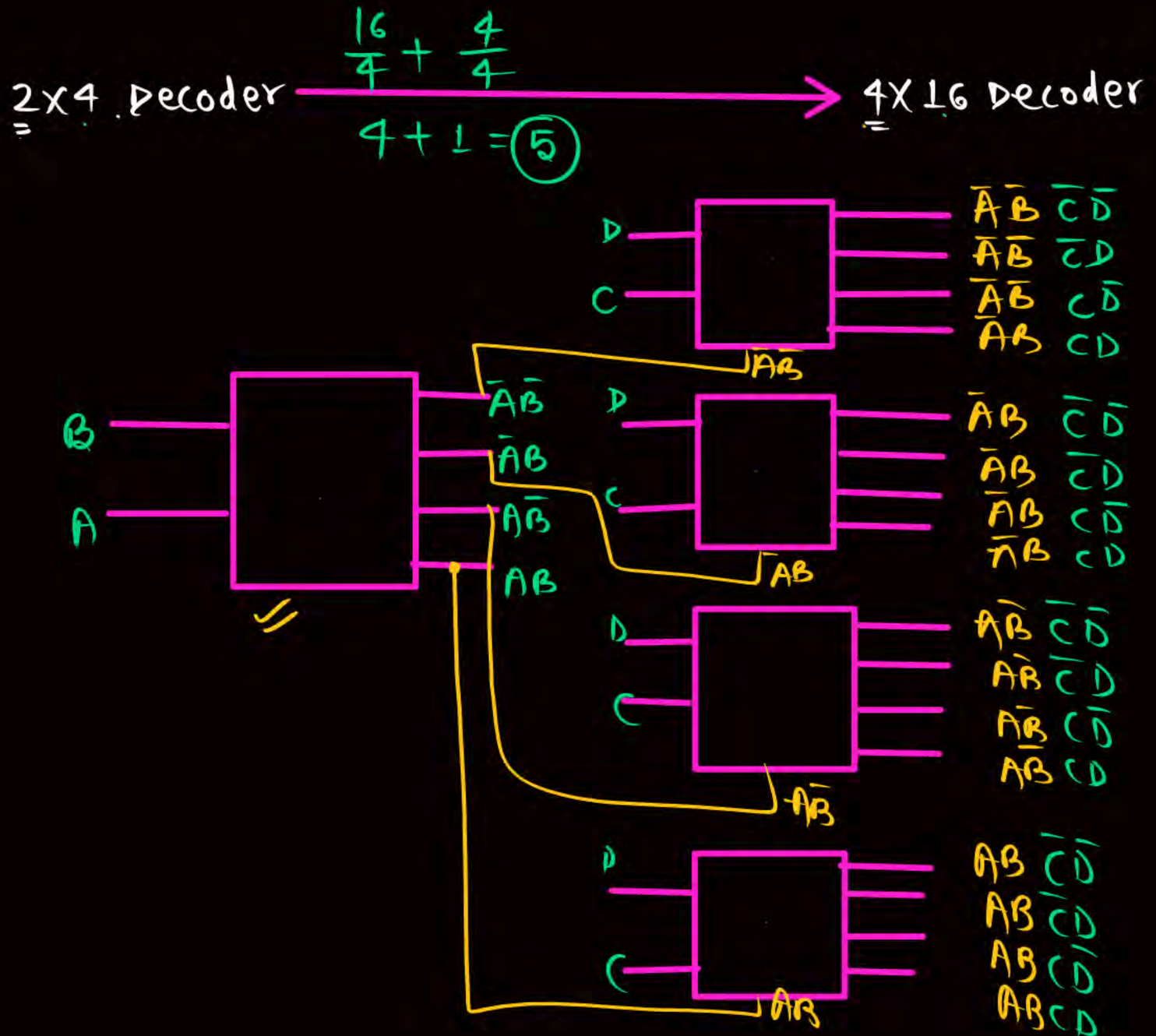
8 output




C
B
A
0
1
2
3
$\bar{A}\bar{B}\bar{C} = y_0$
$\bar{A}\bar{B}C = y_1$
$\bar{A}B\bar{C} = y_2$
$\bar{A}BC = y_3$
$\bar{A}$
$A\bar{B}\bar{C} = y_4$
$A\bar{B}C = y_5$
$AB\bar{C} = y_6$
$ABC = y_7$
A


8/4
2x
4
3x
8
double

Ans = 2


2x4 decoder
16/4 + 4/4
4 + 1 = 5
4x16 decoder
B
A
D
C
$\bar{A}\bar{B}$
$\bar{A}B$
$A\bar{B}$
AB
$\bar{A}\bar{B}\bar{C}\bar{D}$
$\bar{A}\bar{B}\bar{C}D$
$\bar{A}\bar{B}C\bar{D}$
$\bar{A}\bar{B}CD$
$\bar{A}B\bar{C}\bar{D}$
$\bar{A}B\bar{C}D$
$\bar{A}BC\bar{D}$
$\bar{A}BCD$
$A\bar{B}\bar{C}\bar{D}$
$A\bar{B}\bar{C}D$
$A\bar{B}C\bar{D}$
$A\bar{B}CD$
$AB\bar{C}\bar{D}$
$AB\bar{C}D$
$ABC\bar{D}$
$ABCD$

Q 3×8 Decoder $\xrightarrow{\frac{64}{8} + \frac{8}{8}}$ 6×64 Decoder

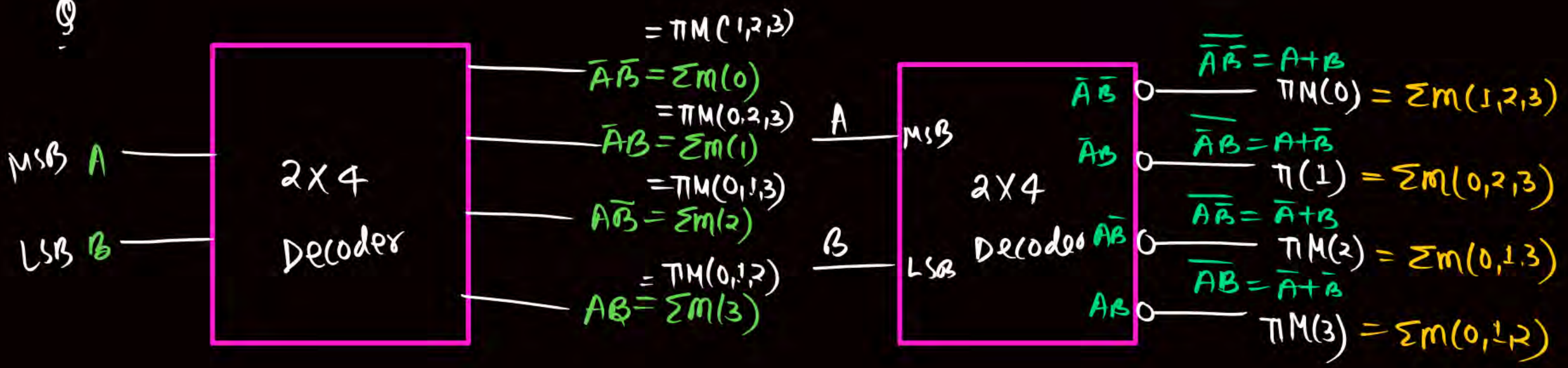
$8 + 1 = 9$

$2 \times 4 \xrightarrow{\frac{8}{4} + \frac{2}{4}} 3 \times 8$ Decode

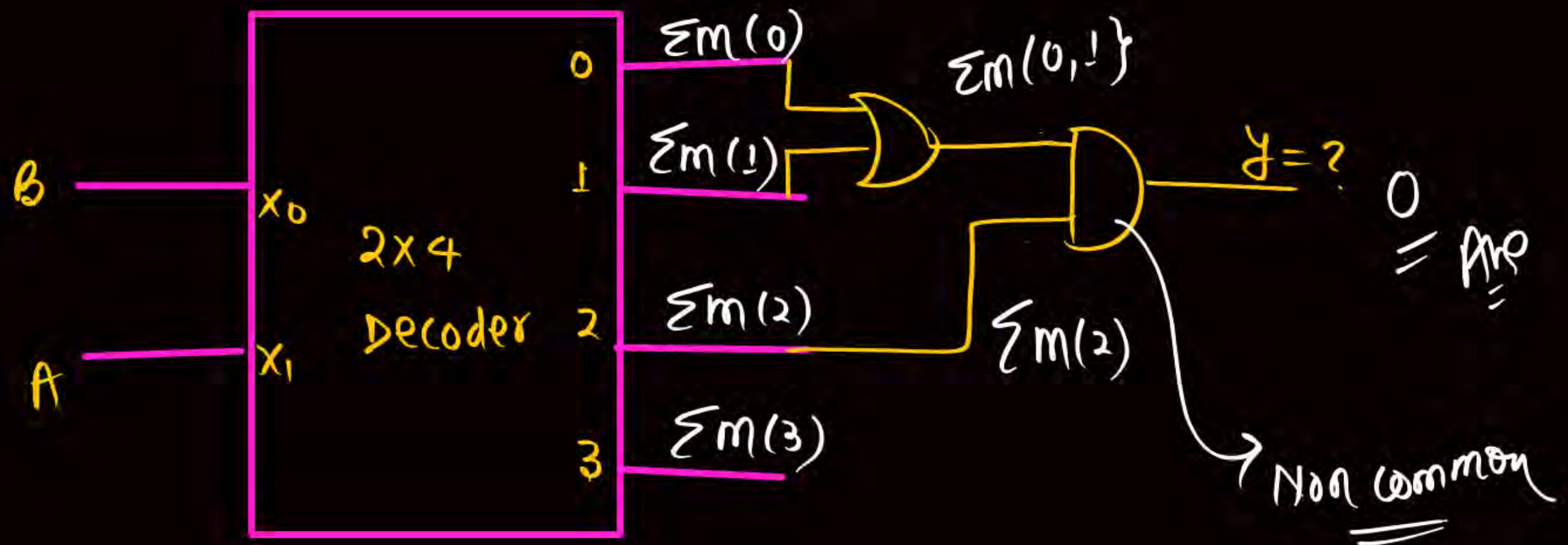
2 AND

—

Q

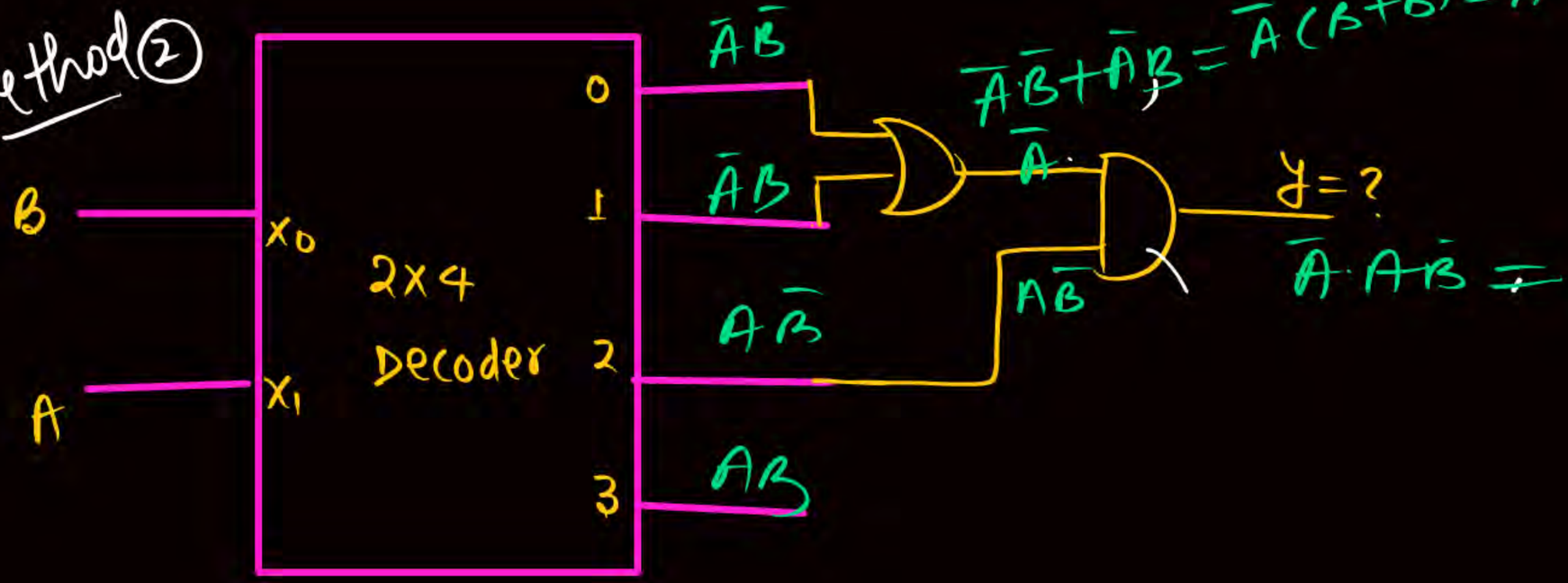

MSB A
LSB B
2X4 Decoder
$\bar{A}\bar{B} = \Sigma m(0) = \pi M(1,2,3)$
$\bar{A}B = \Sigma m(1) = \pi M(0,2,3)$
$A\bar{B} = \Sigma m(2) = \pi M(0,1,3)$
$AB = \Sigma m(3) = \pi M(0,1,2)$
A
B
MSB
LSB
2X4 Decoder
$\bar{A}\bar{B}$
$\bar{A}B$
$A\bar{B}$
AB
$\overline{\bar{A}\bar{B}} = A+B$
$\pi M(0) = \Sigma m(1,2,3)$
$\overline{\bar{A}B} = A+\bar{B}$
$\pi(1) = \Sigma m(0,2,3)$
$\overline{A\bar{B}} = \bar{A}+B$
$\pi M(2) = \Sigma m(0,1,3)$
$\overline{AB} = \bar{A}+\bar{B}$
$\pi M(3) = \Sigma m(0,1,2)$

Q


B
A
x_0
x_1
2x4
Decoder
0
1
2
3
$\Sigma m(0)$
$\Sigma m(1)$
$\Sigma m(2)$
$\Sigma m(3)$
$\Sigma m(0,1)$
$\Sigma m(2)$
Y = ?
0
Non common

Q.

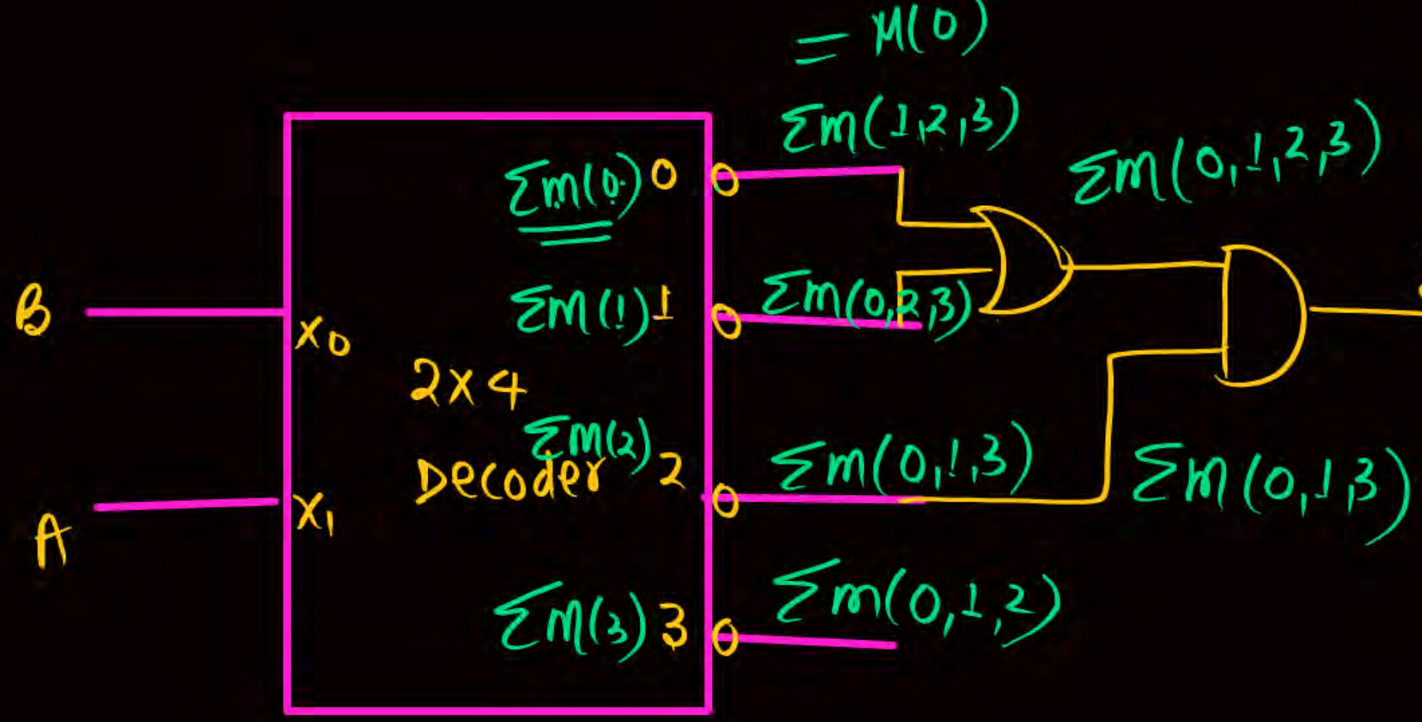
Method ②



Ans

1.0

Q.



$\bar{A}+B$

$y = \Sigma m(0,1,3) = \Pi M(2)$

$= \bar{A}\bar{B} + \bar{A}B + AB$

$= \bar{A}(\bar{B}+B) + AB$

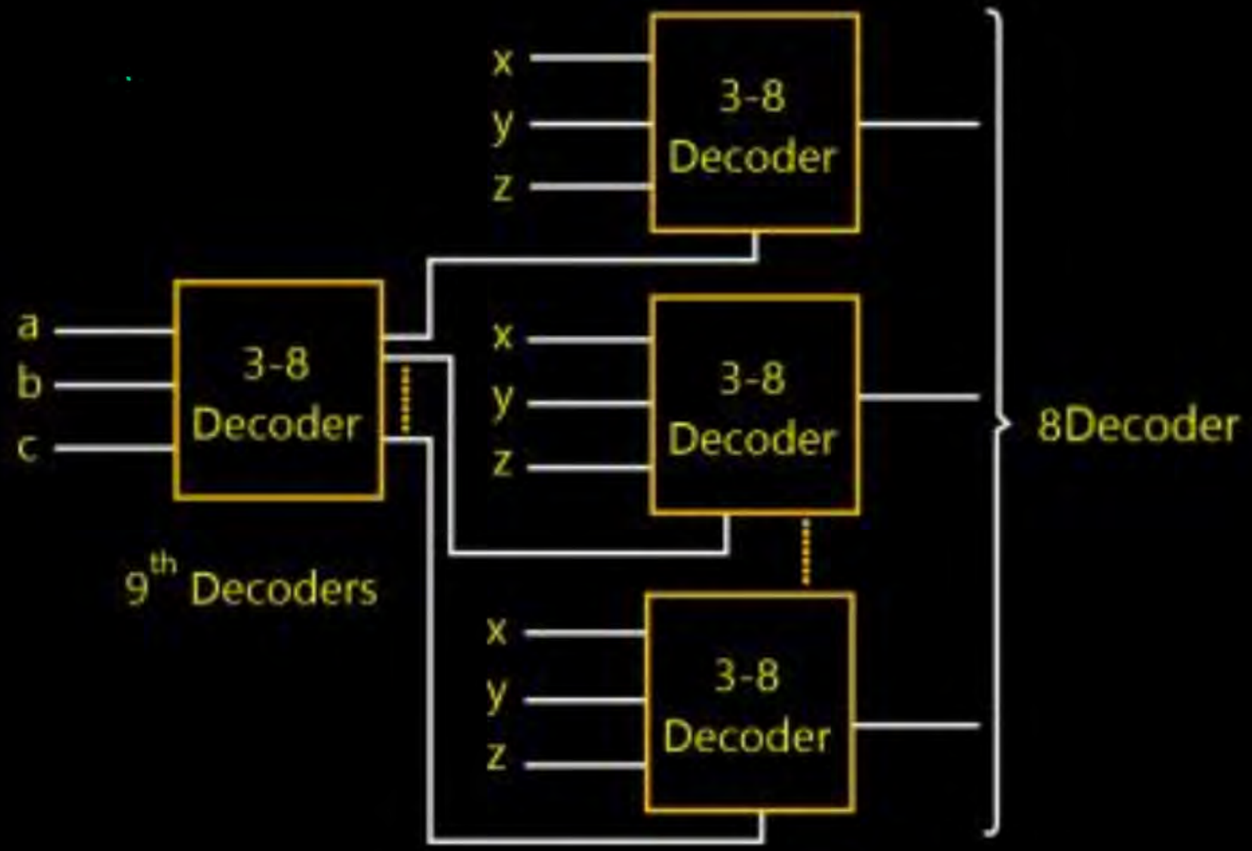
$= \bar{A} + AB$

$= (\bar{A}+A)(\bar{A}+B)$

$= \bar{A}+B$ Ans

DECODER

DESIGN 6 × 64 DECODER BY USING 3 × 8 DECODER



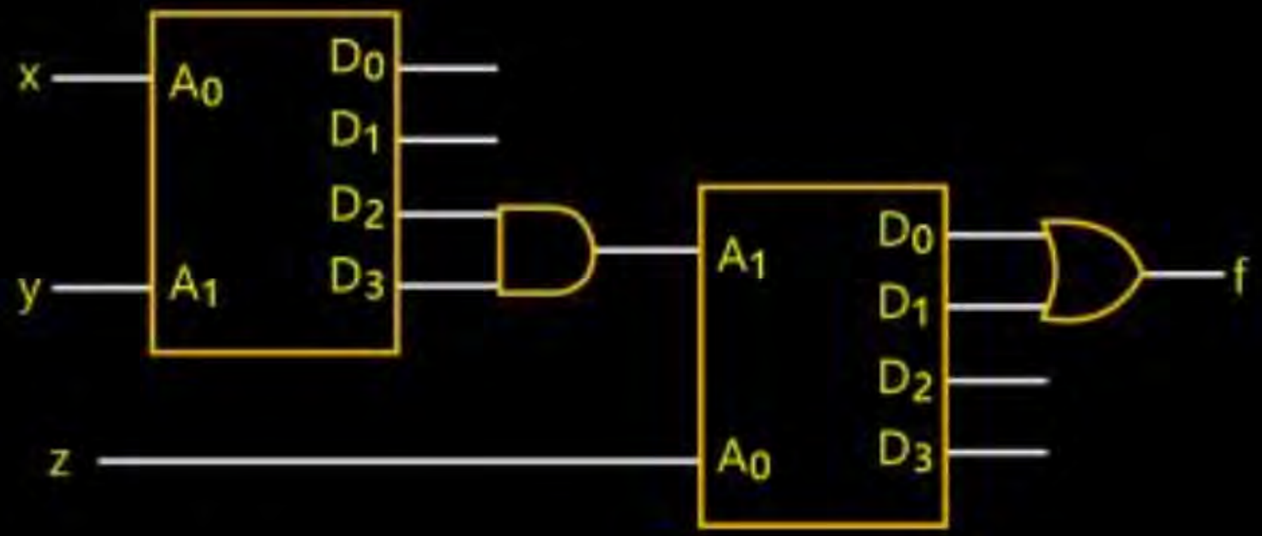
Question **(MCQ)**

★★☆

A logic circuit consist of two 2 × 4 decoders as shown in the figure. The output of decoder are as follow:

D0 = 1 when A0 = 0, A1 = 0
D1 = 1 when A0 = 1, A1 = 0
D2 = 1 when A0 = 0, A1 = 1
D3 = 1 when A0 = 1, A1 = 1

The value of f(x, y, z) is



A 0

B z

C $\bar{z}$

D 1

Question **(MCQ)**

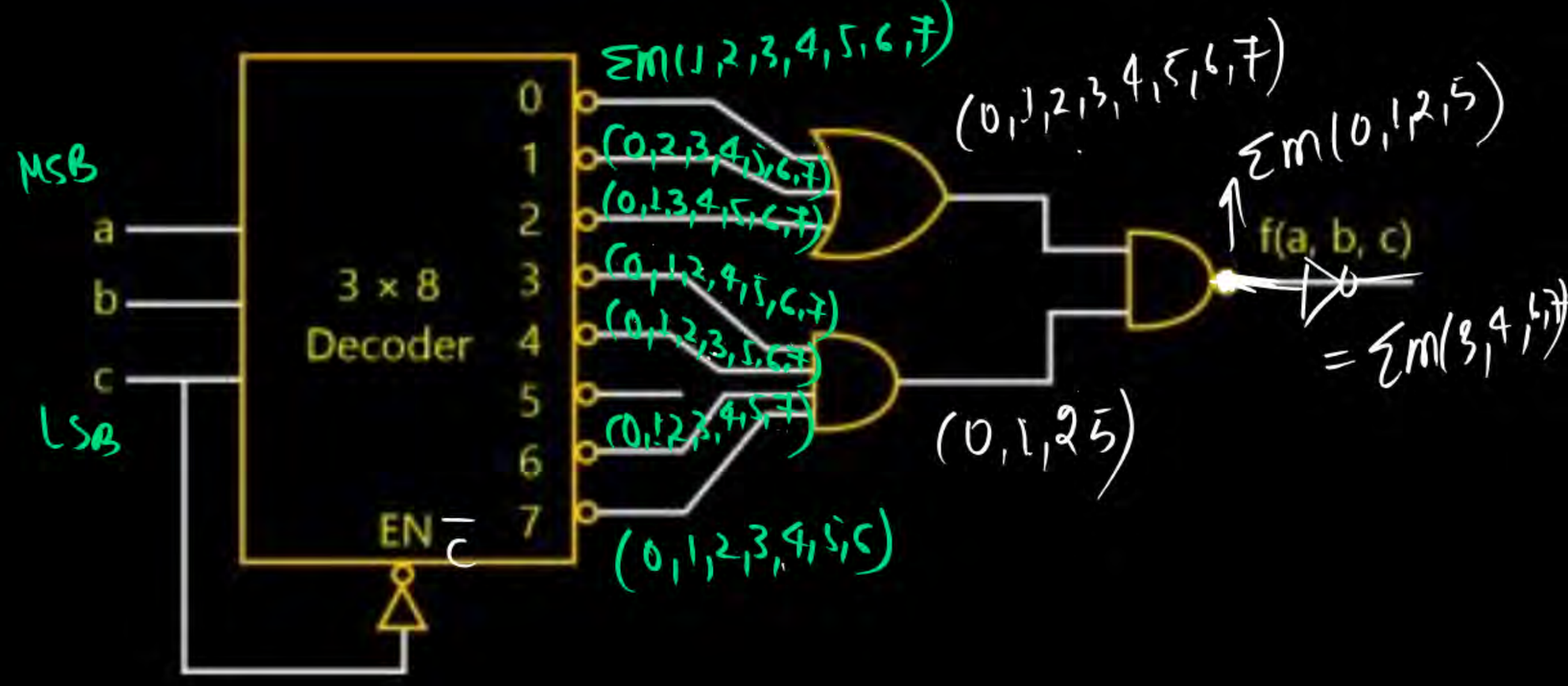
The Boolean expression f(a, b, c) in its canonical form for the decoder circuit shown below is

A $\Pi M(4, 6)$

B $\Sigma m(0, 1, 2, 3, 5, 7)$

C $\Sigma m(4, 6)$

D $\Pi M(0, 1, 2, 3, 5)$



$$y = \bar{C}\ \Sigma m(3,4,6,7)$$

$$= \bar{C}\left[\bar{A}BC + A\bar{B}\bar{C} + AB\bar{C} + ABC\right]$$

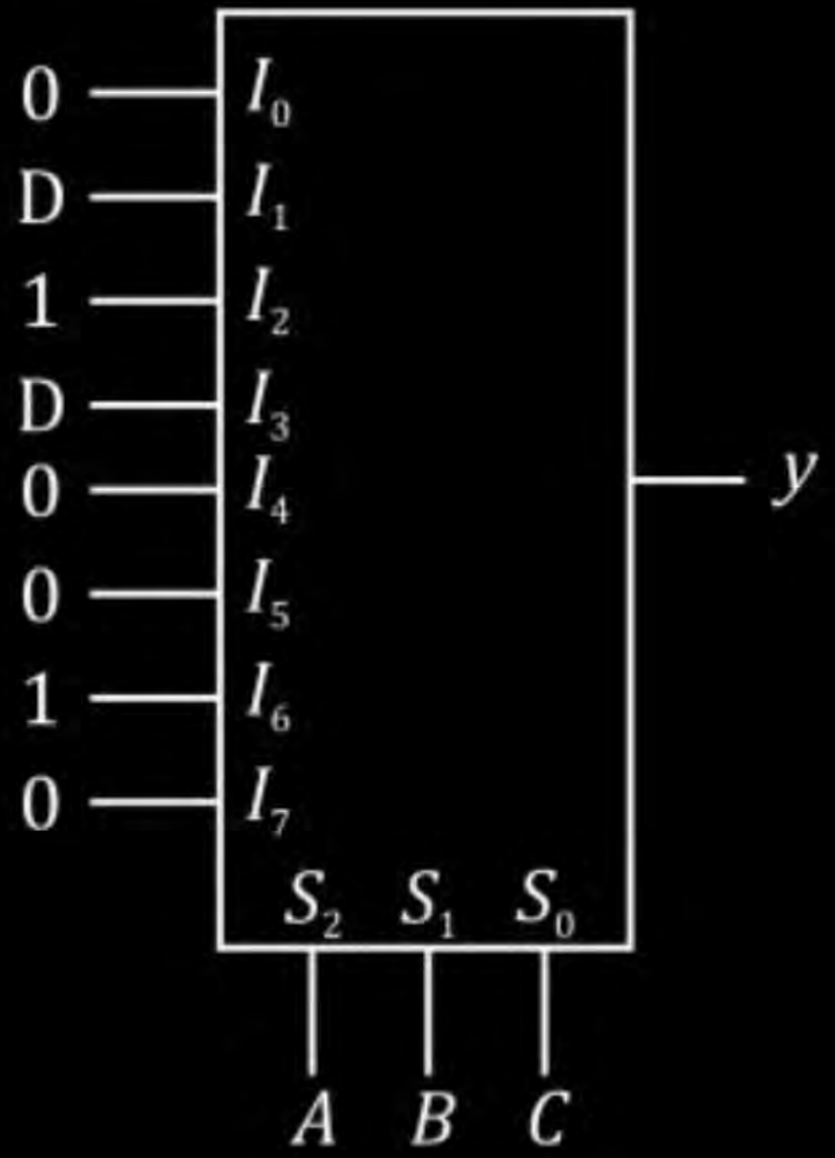
$$= A\bar{B}\bar{C} + AB\bar{C}$$

$$= \Sigma m(4,6)\ \checkmark$$

$$= \Pi M(0,1,2,3,5,7)$$

Question **(MCQ)**

A digital circuit is as given below :



$y\,(A, B, C, D)$ will be

A $A\bar{B}C + A\bar{C}D$

B $BC + A\bar{B}D$

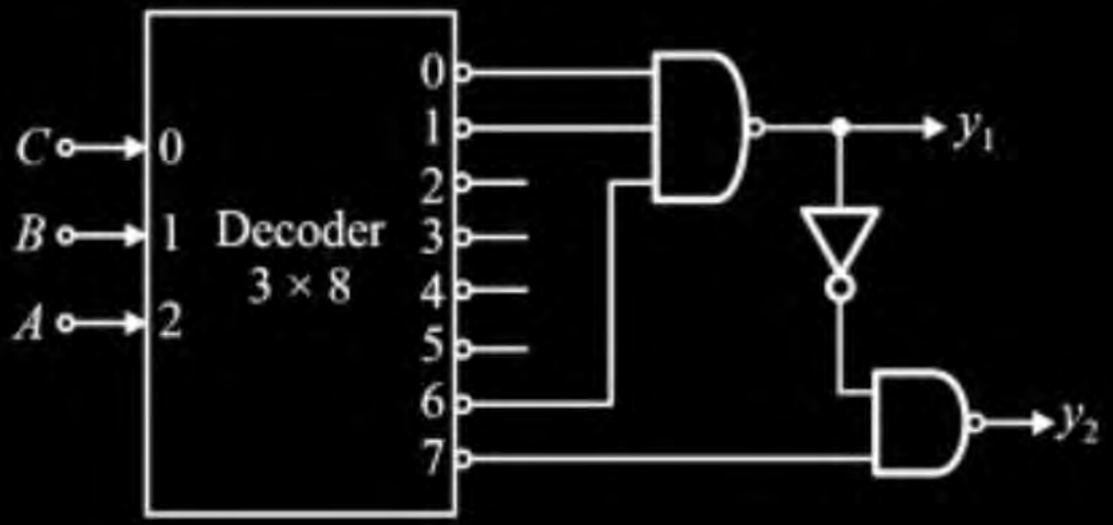
C $B\bar{C} + \bar{A}CD$

D $\bar{B}C + A\bar{B}D$

Question [MCQ]

★★☆

A 3 line to 8 line decoder with three inputs A, B, C and two outputs y_1 and y_2, is configured as shown below. The minimized expression of outputs will be



A $y_1 = \bar{A}\bar{B} + AB\bar{C};\ y_2 = \overline{A \oplus B}$

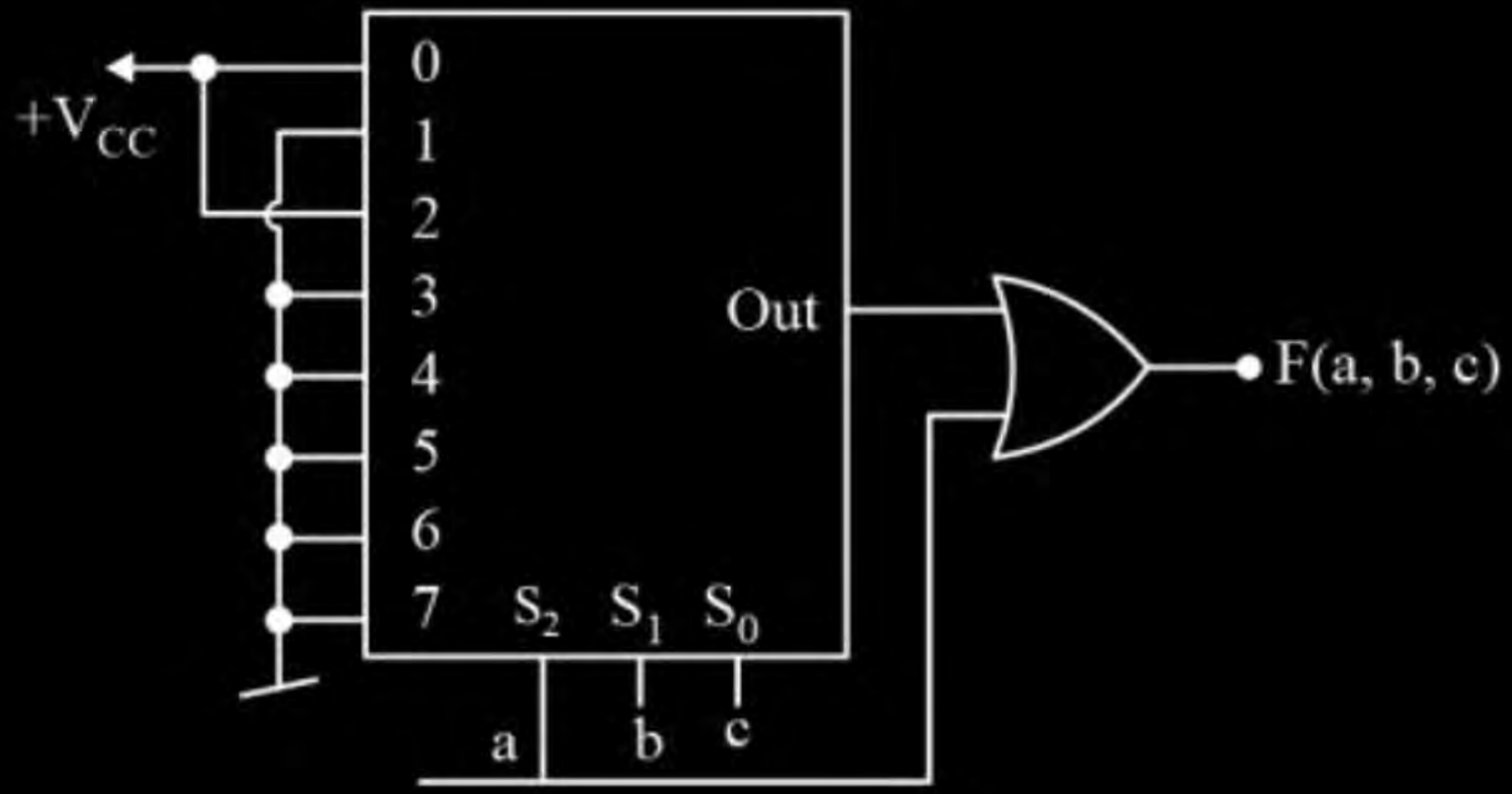
B $y_1 = AB + \bar{A}\bar{B}C;\ y_2 = A \oplus B$

C $y_1 = \bar{A}B + A\bar{C};\ y_2 = AB + AC$

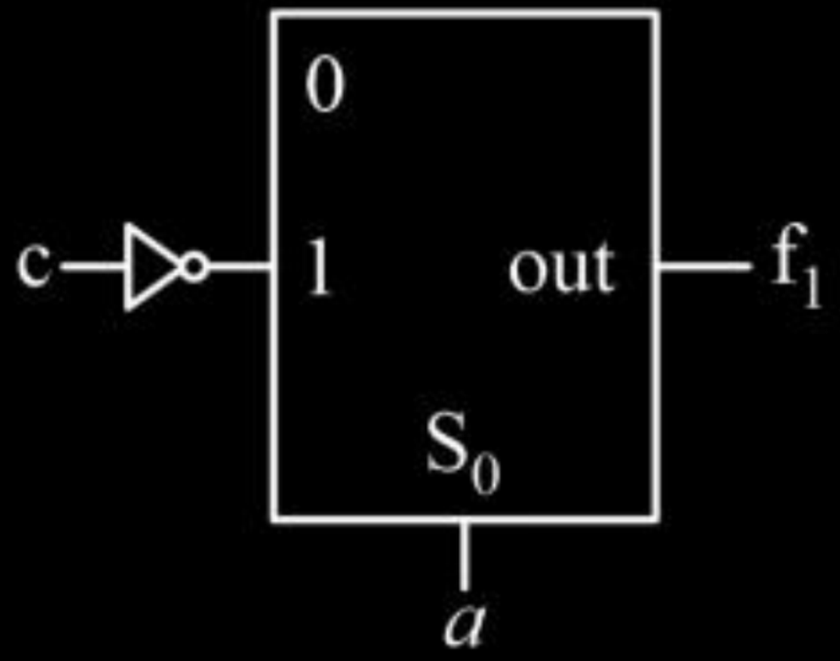
D $y_1 = A\bar{B} + \bar{A}C;\ y_2 = \bar{A}B + \bar{B}C$

Question [MCQ]

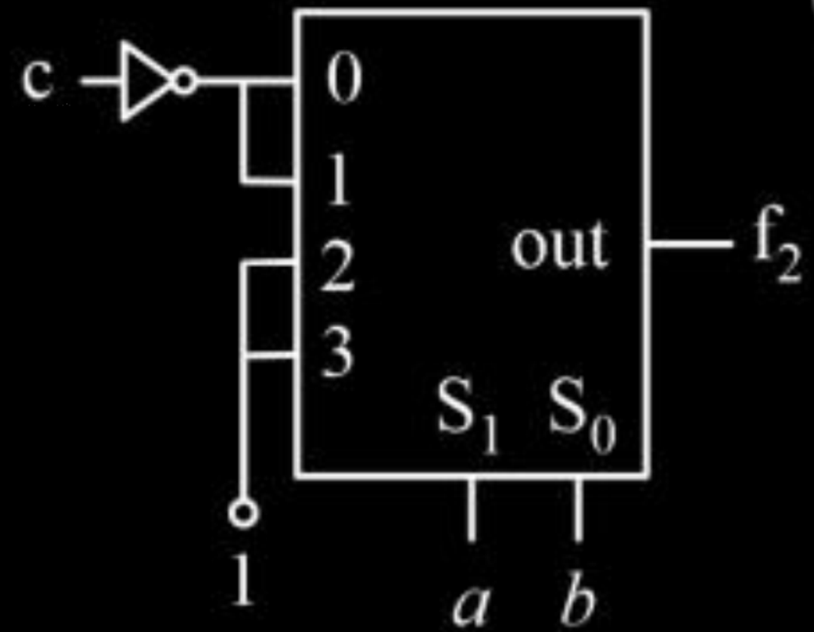
An 8 ×1 [MUX is configured as shown below to implement a function f(a, b, c), a is MSB. Alternatively, the same function can also be implemented using smaller size 2 × 1 MUX and 4 × 1 MUX. The correct alternative is



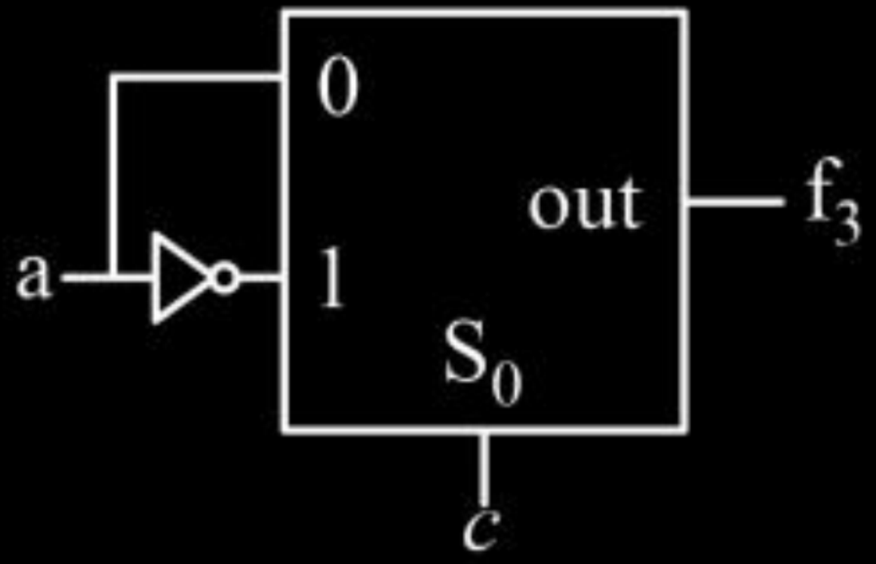
A


0
1
out
c
f_1
S_0
a

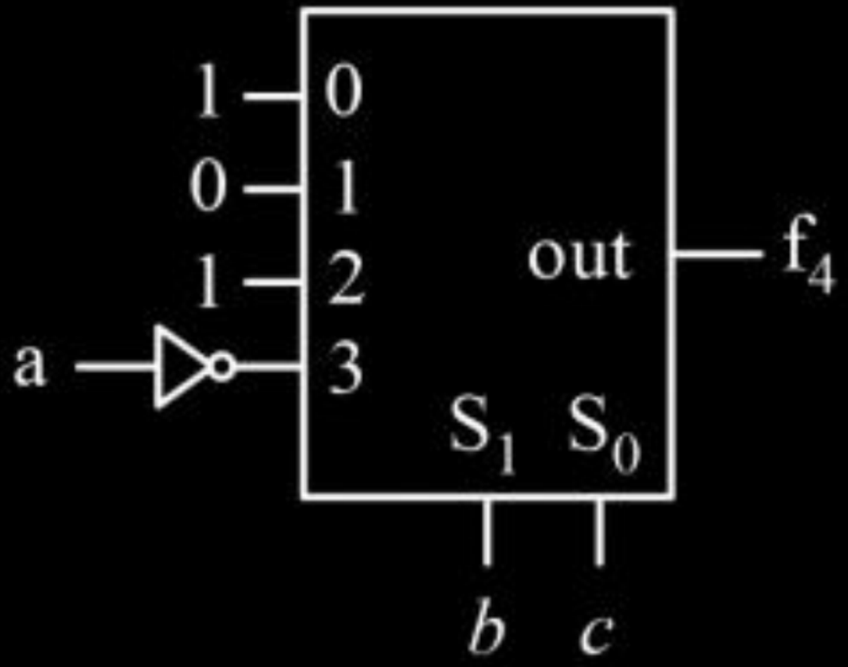
B


c
0
1
2
3
out
f_2
S_1 S_0
1
a b

C


0
out
f_3
a
1
S_0
c

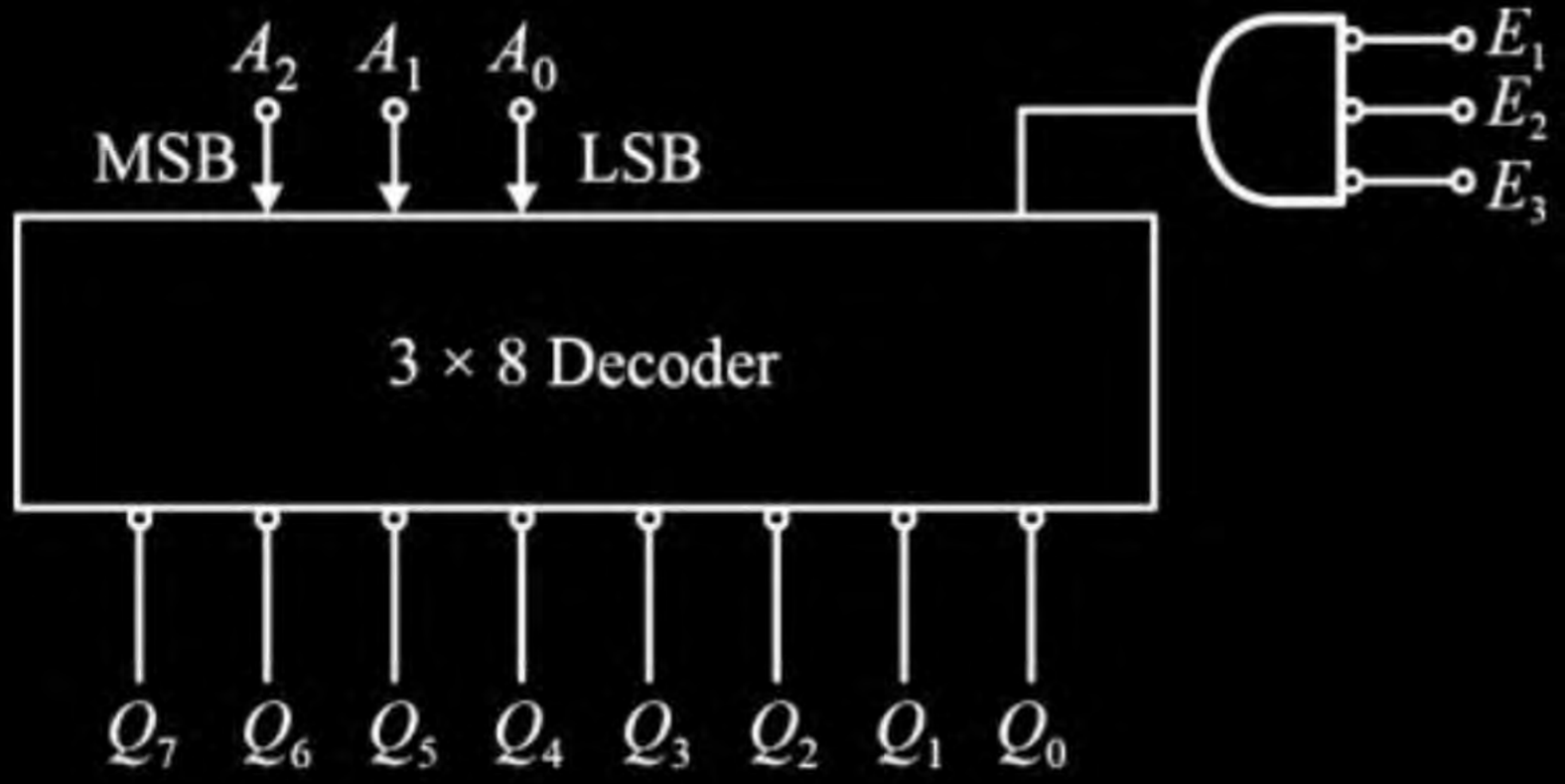
D


1
0
1
0
1
2
3
out
f_4
a
S_1 S_0
b c

Question [MCQ]

The logic diagram of a 3 × 8 decoder with active low outputs is shown below.

What is state of outputs Q_7, ... Q_0 for the set of inputs $E_3 = E_1 = 1$, $E_2 = 0$, $A_2 = A_1 = 1$ and $A_0 = 0$?



A 1111 1111

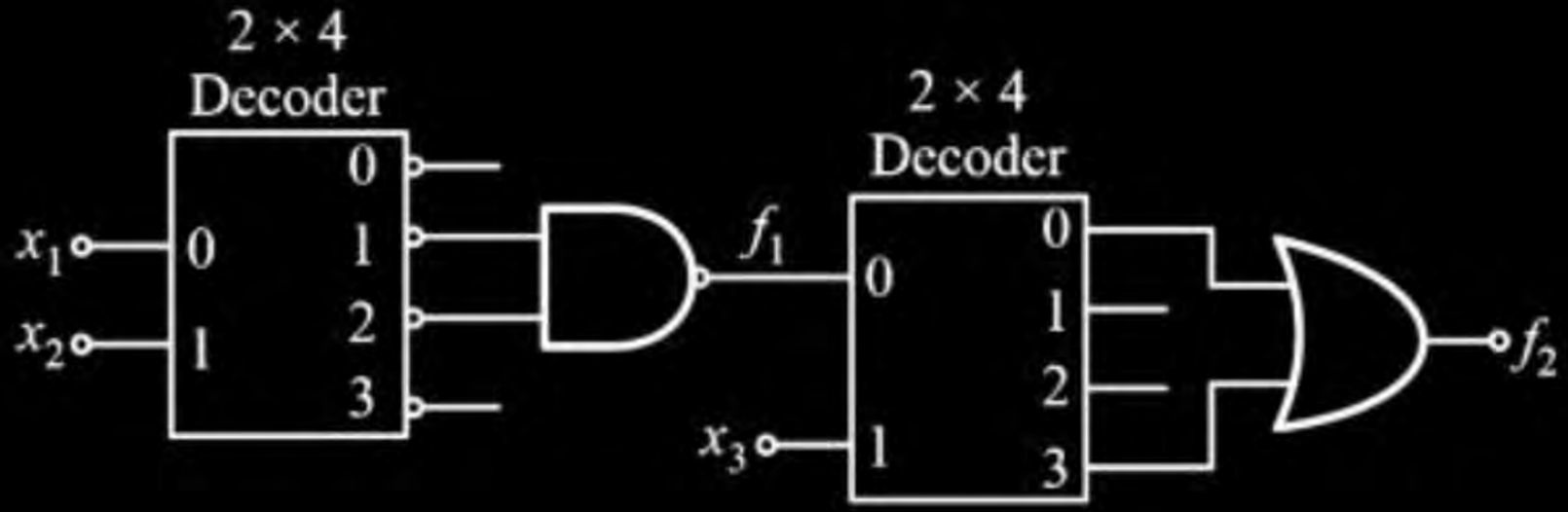
B 1011 1111

C 1111 0111

D 0000 0000

Question [MCQ]

Two 2 × 4 decoders one with active low outputs and another with active high output are interconnected as shown below. The output function $f_2(x_3, x_2, x_1)$ will be



A $f_2 = (x_1 \oplus x_2) \odot x_3$

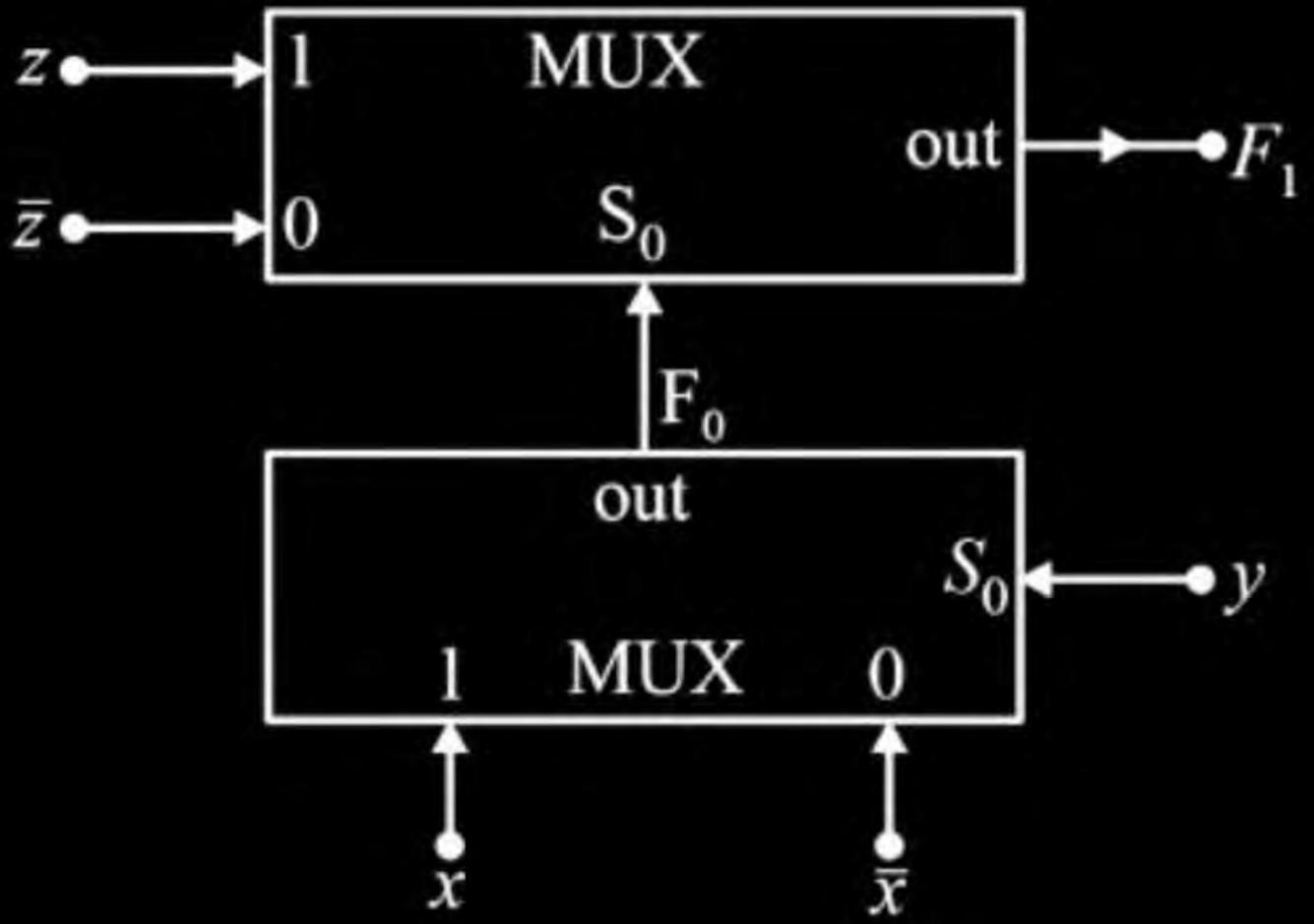
B $f_2 = (x_1 \odot x_2) \odot x_3$

C $f_2 = (x_1 \oplus x_2) \oplus x_3$

D $f_2 = (x_1 \oplus x_2) \oplus x_3$

Question [MCQ]

In the multiplexer configuration shown below, the output F_1 will be



A $\overline{(x \oplus y)} \oplus z$

B $\overline{\overline{(x \oplus y)} + z}$

C $(x \oplus y) \oplus \bar{z}$

D $(x \oplus y) \oplus z$

Thank you DOSTO